

Abdelrahman Khalifa

Systems Software Engineer — Rust · C · Linux · GSoC 2026 (Linux Foundation)

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

SUMMARY

Systems software engineer working in Rust, C and modern C++ on Linux. Completed Google Summer of Code 2026 with The Linux Foundation (OpenPrinting), building the COSMIC desktop's printer stack and landing 7 PRs upstream in cups-rs and libcups. Comfortable with FFI, IPC, async code and maintainer code review.

OPEN SOURCE

Google Summer of Code 2026 Contributor — The Linux Foundation / OpenPrinting | May 2026 – Oct 2026

COSMIC desktop printer setup tool in Rust; design discussions, code review and documentation with OpenPrinting and System76 mentors

| github.com/Abd002/GSOC-2026 | github.com/Abd002/cosmic-printers

- Designed a printer-management stack as 5 Rust crates: printers-core (types, grouping queues into physical devices), -client (async varlink client), -server (CUPS, IPP, DNS-SD and Printer Application backend), -ui (iced/libcosmic) and -app.
- Exposed the backend over varlink IPC from cosmic-settings-daemon, so the C FFI code never loads in the Settings GUI process; the standalone app falls back to running the server in-process.
- Built a Printers page and an Add Printer flow for COSMIC Settings: live updates, print queue control, supply levels, and legacy printers set up through Printer Applications (PAPPL) instead of PPD drivers.
- Tested on real HP printers and Printer Applications (Ghostscript, Gutenprint, HPLIP); CI checks formatting, Clippy and tests on several architectures and runs CUPS, Avahi and ippeveprinter to test discovery.

Upstream contributions — OpenPrinting cups-rs (Rust) and libcups (C) | Jan 2026 – Oct 2026

- cups-rs: ported to libcups3 with CUPS 2 and CUPS 3 builds and CI; added IPP management operations, Printer Application support, connections by IP/port, lpoptions support and FFI bindings for the libcups3 DNS-SD API (PRs #18–#22, merged).
- libcups: fixed crashes in cupsDNSSDDelete() and a leaked client in the Avahi DNS-SD code (#164, #165, merged).
- COSMIC: Printers page in Settings (cosmic-settings #2182) and the varlink printers server (cosmic-settings-daemon #188), in review.
- Winter of Code 2026: implemented the lp and lpstat commands in Rust on cups-rs (PR #11).

Contributor — Boa JavaScript engine (Rust), Rust Clippy, Zephyr RTOS | Mar 2024 – Dec 2025

- Boa: optimized JsStr::code_points() for Latin1 strings, avoiding UTF-16 conversion; with tests (PR #4553).
- Rust Clippy: fixed ref_as_ptr false positives for non-temporary references in let/const (PR #16214).
- Zephyr RTOS: implemented the IP_MULTICAST_LOOP socket option across 6 networking modules, merged upstream (commit b117036).

EXPERIENCE

Embedded Systems Engineer — EKSON | Aug 2025 – Present

- Built a modern C++/Qt control application for motion simulators and optimized its performance by ~40% through asynchronous processing and fewer threads; reliable UART control.
- Developed Zephyr RTOS firmware for an ESP32-S3 3-DoF simulator, migrating from FreeRTOS for more deterministic behavior and production readiness.

Embedded Linux Systems Intern — ODC | Dec 2023 – Jan 2024

- Built Linux systems with Yocto and Buildroot; worked with kernel modules, device drivers, device trees, file systems and hardware flashing.

TECHNICAL SKILLS

Languages: Rust, C, C++ (modern C++, Qt), Python, Bash, Java

Systems & Linux: Linux (Ubuntu/Debian), operating systems, FFI and bindings, IPC (varlink, D-Bus; zbus, zlink), concurrency (async/event-driven, multithreading), memory management, networking (TCP/UDP), CUPS/IPP, DNS-SD/mDNS (Avahi), kernel modules and device drivers

Tooling: Cargo, Clippy, rustfmt, Git/GitHub, CI, CMake, Make, Yocto, Buildroot, QEMU, testing, debugging, performance optimization, code review, documentation

Desktop & embedded: iced, libcosmic, Qt; Zephyr, FreeRTOS, ARM Cortex-M, ESP32

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa) and 4-time ECPC finalist (Egyptian Collegiate Programming Contest).
- Deloitte Software Engineering Mentorship (React Native, 2024); IEEE student chapter Treasurer (\$10k+ budget).

EDUCATION

B.Sc. Computer and Systems Engineering — Minia University | 2020 – 2025

Very Good (84.6%) with Honors. Graduation project: AUTOSAR-based V2X vehicle prototype.